

Figures shown in the right margin indicate the full marks for each question.

Answer Six questions, taking Three from each section including Q.1, Q.2, Q.5 and Q.6.

Use separate answer script for each section.

Assume reasonable data if necessary.

Section-A

Marks

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|-----|--|----|
| Q.1 | Write down the working rules to solve the first order exact ordinary differential equations. Identify the differential equation and solve it with suitable method: | 10 |
| | (i) $(x^2 - 4xy - 2y^2)dx + (y^2 - 4xy - 2x^2)dy = 0$ | |
| | (ii) $\frac{dy}{dx} = \sin(x+y) + \cos(x+y)$. [CO1, PO1, C1] | |
| Q.2 | (a) Write down the general form of reducible to linear ordinary differential equations of order one. Solve the following differential equation: $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$.
[CO1, PO1, C3] | 4 |
| | (b) Estimate the solution of the following first order but not first-degree ordinary differential equations: $y = 2px + yp^2$ where $p = \frac{dy}{dx}$. [CO1, PO1, C3] | 3 |
| | (c) Let population of country be decreasing at the rate proportional to its population. If the population has decreased to 25% in 10 years, how long will it take to be half?
[CO3, PO2, C3] | 3 |
| Q.3 | Obtain the solution of the following ordinary differential equations (Any three including (i)): [CO1, PO1, C3] | 10 |
| | (i) $\frac{d^4y}{dx^4} + y = 0$, (ii) $(D+2)(D-1)^3 y = e^x$, | |
| | (iii) $(D^2 - 3D + 2)y = \sin 3x$, (iv) $(D^2 - 2D + 1)y = e^{3x}x^2 + \cos 7x$. | |
| Q.4 | (a) What do you mean by initial value problem? Solve the IVP:
$y'' - 4y = e^x$; $y(0) = 1$, $y'(0) = 0$. [CO3, PO2, C3] | 4 |
| | (b) State boundary value problem. Determine the solution of the following boundary value problem: $y'' + 9y = \cos x$; $y'(0) = 5$, $y(\pi/2) = -5/3$. [CO3, PO2, C3] | 6 |

Section-B

Marks

2

Q.5 (a) Define each of the terms with an example “semi-linear and quasi-linear partial differential equation of order one”. [CO2, PO1, C1]

(b) Construct a partial differential equation by eliminating arbitrary constants h , and k from the equation $(x-h)^2 + (y-k)^2 + z^2 = \lambda^2$, hence mention the order and degree of the obtained equation. [CO2, PO1, C2]

(c) Obtain the solution of the Lagrange’s equation: $p \tan x + q \tan y = \tan z$. [CO2, PO1, C3]

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Q.6 (a) Write down the Charpit’s auxiliary equation. Determine the complete integral of the following first order non-linear partial differential equations (any one):

(i) $3p^2 - q = 0$, (ii) $(p^2 z^2 + q^2) z^2 = 1$. [CO2, PO1, C3]

(b) Solve the following homogeneous linear partial differential equations (any one):

(i) $r - 2s + t = \sin(2x + 3y)$,

(ii) $(D^3 - 6D^2 D' + 11DD'^2 - 6D'^3)z = e^{5x+6y}$. [CO2, PO1, C3]

5

Q.7 (a) Define central tendency. Give a short discussion on the classification of central tendency. [CO3, PO1, C2]

(b) What do you understand by standard deviation? Calculate the geometric mean and standard deviation from the following data: [CO3, PO2, C3]

3

7

Class	0 - 6	6 - 12	12 - 18	18 - 24	24 - 30
Frequency	8	10	12	9	5

Q.8 (a) Define probability mass function. Suppose, a fair coin tosses three times. Draw the probability mass function. Also determine the probability that at least one head is flipped. [CO3, PO2, C1]

(b) State the conditional probability. Suppose that a fair coin is tossed three times. Given that exactly two of the coin tosses were tails, what is the probability that the second flip was head? [CO3, PO2, C2]

(c) Suppose that 15% of visitors to a website are from the USA, 35% are from Australia, and 50% from the rest of the world. The probabilities that a visitor from that region purchases something are 0.01, 0.05 and 0.02 respectively. Now, find the probability that a visitor purchases something. [CO3, PO2, C3]

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3